

React JS Hands-On Lab

React Forms – ticketraisingapp

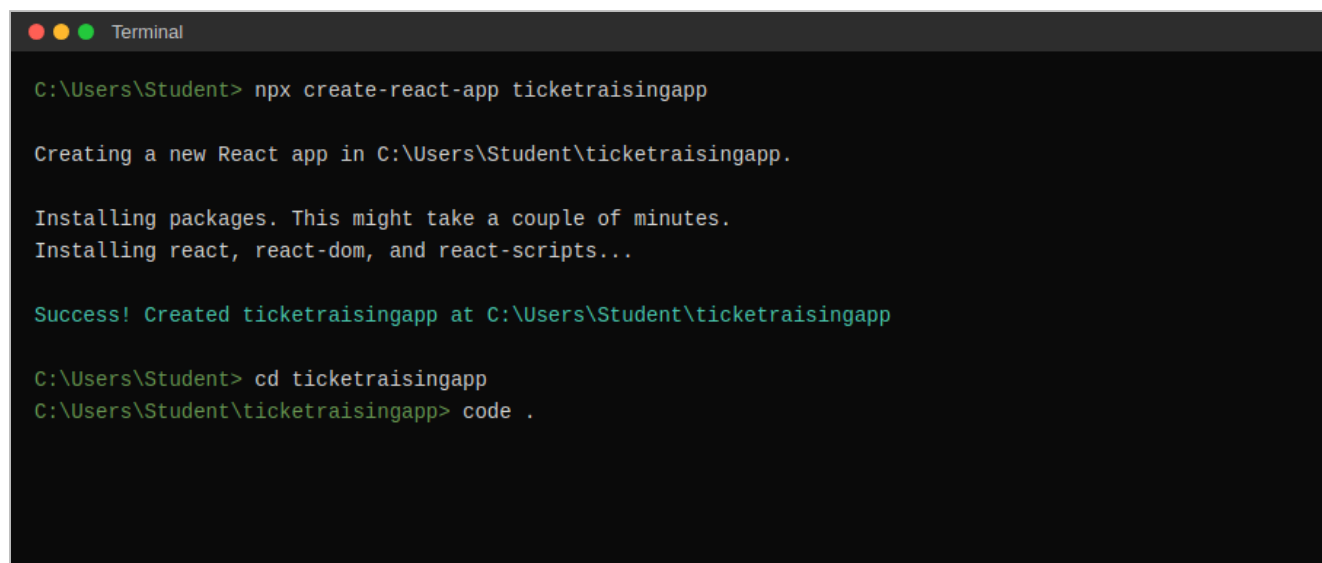
Objectives

- Explain about React forms
- Define controlled components
- Explain about various input controls
- Explain about handling forms
- Explain about submitting forms

In this lab I implemented a React form using controlled input controls like textbox and textarea.

Setup: Create the React app "ticketraisingapp"

Command used: `npx create-react-app ticketraisingapp`



```
Terminal
C:\Users\Student> npx create-react-app ticketraisingapp

Creating a new React app in C:\Users\Student\ticketraisingapp.

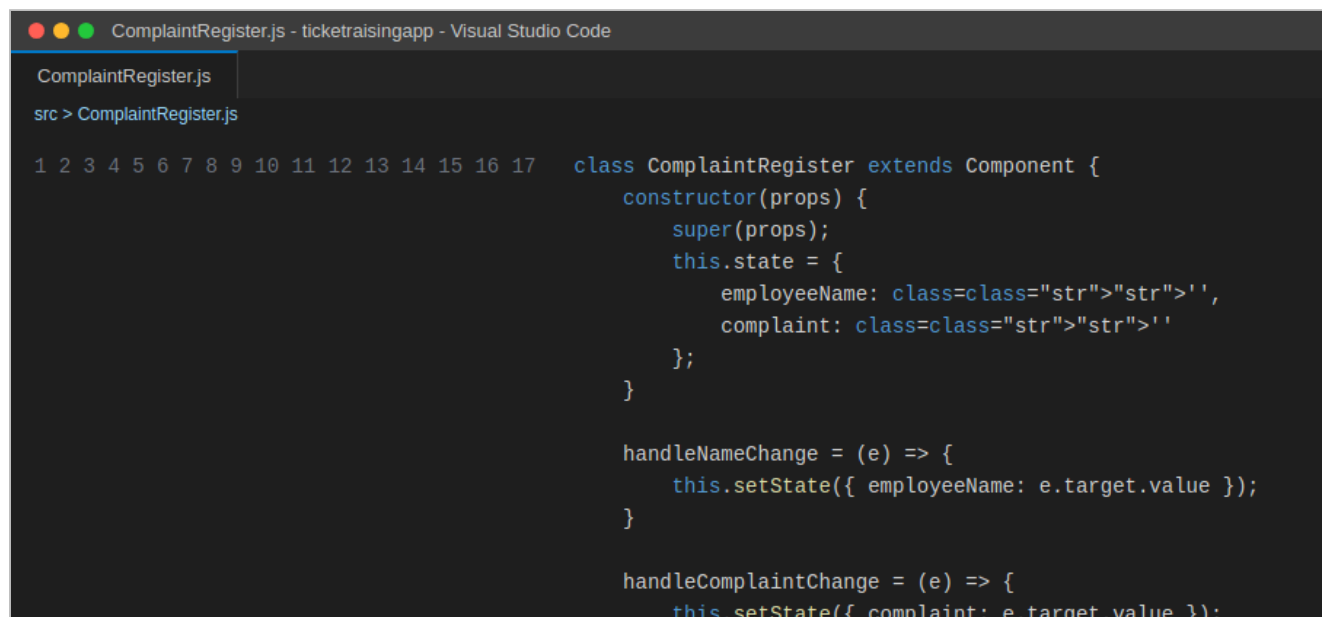
Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts...

Success! Created ticketraisingapp at C:\Users\Student\ticketraisingapp

C:\Users\Student> cd ticketraisingapp
C:\Users\Student\ticketraisingapp> code .
```

Step 1: ComplaintRegister component – state and change handlers

A textbox for employee name and a textarea for the complaint are made into controlled components using state.



```
ComplaintRegister.js - ticketraisingapp - Visual Studio Code
src > ComplaintRegister.js

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17  class ComplaintRegister extends Component {
                                     constructor(props) {
                                         super(props);
                                         this.state = {
                                             employeeName: class=class="str">"str">'',
                                             complaint: class=class="str">"str">'
                                         };
                                     }

                                     handleNameChange = (e) => {
                                         this.setState({ employeeName: e.target.value });
                                     }

                                     handleComplaintChange = (e) => {
                                         this.setState({ complaint: e.target.value });
                                     }
                                     }
```

Step 2: handleSubmit event of the button

On submit, a reference number is generated and shown in an alert box for further follow up.

```
ComplaintRegister.js - ticketraisingapp - Visual Studio Code
ComplaintRegister.js
src > ComplaintRegister.js

1      handleSubmit = (e) => {
2          e.preventDefault();
3          const referenceNumber = Math.floor(100000 + Math.random() * 900000);
4          alert(class="str">"Complaint registered successfully!\nReference Number: " + referenceNumber)
5      }
```

Step 3: render() – the complaint form

```
ComplaintRegister.js - ticketraisingapp - Visual Studio Code
ComplaintRegister.js
src > ComplaintRegister.js

1 2 3 4 5 6      render() {
7 8 9 10 11      return (
12 13 14 15          <class="tag">form onSubmit={this.handleSubmit}>
16 17 18 19              <class="tag">h2>Complaint Register</class="tag">h2>
20 21 22 23
24 25 26              <class="tag">label>Employee Name: </class="tag">label>
                  <class="tag">input
                      type=class="str">"text"
                      value={this.state.employeeName}
                      onChange={this.handleNameChange}
                  />
                  <class="tag">br /><class="tag">br />

                  <class="tag">label>Complaint: </class="tag">label>
                  <class="tag">textarea
                      value={this.state.complaint}
                      onChange={this.handleComplaintChange}
                      rows=class="str">"4"
                      cols=class="str">"30"
                  />
                  <class="tag">br /><class="tag">br />

                  <class="tag">button type=class="str">"submit">Submit</class="tag">button>
              </class="tag">form>
          );
      }
```

Step 4: Add ComplaintRegister to App.js

```
App.js - ticketraisingapp - Visual Studio Code
App.js
src > App.js

1 2 3 4 5 6 7      class="kw">import React from class="str">"str">'react';
8 9 10 11 12      class="kw">import ComplaintRegister from class="str">"str">'./ComplaintRegister';
13
      function App() {
          return (
              <class="tag">div className=class="str">"App">
                  <class="tag">ComplaintRegister />
              </class="tag">div>
          );
      }
```

Step 5: Build and run using npm start

```
Terminal
C:\Users\Student\ticketraisingapp> npm start

Compiling...

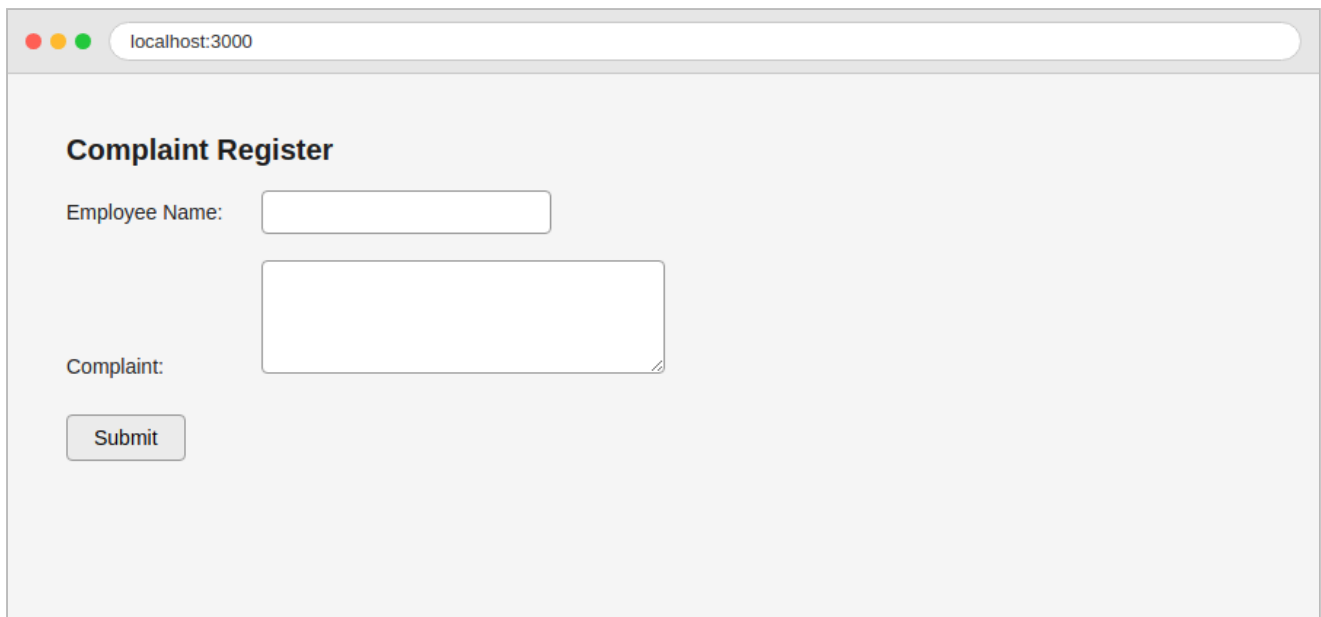
Compiled successfully!

You can now view ticketraisingapp in the browser.

Local:      http://localhost:3000
On Your Network:  http://192.168.1.7:3000

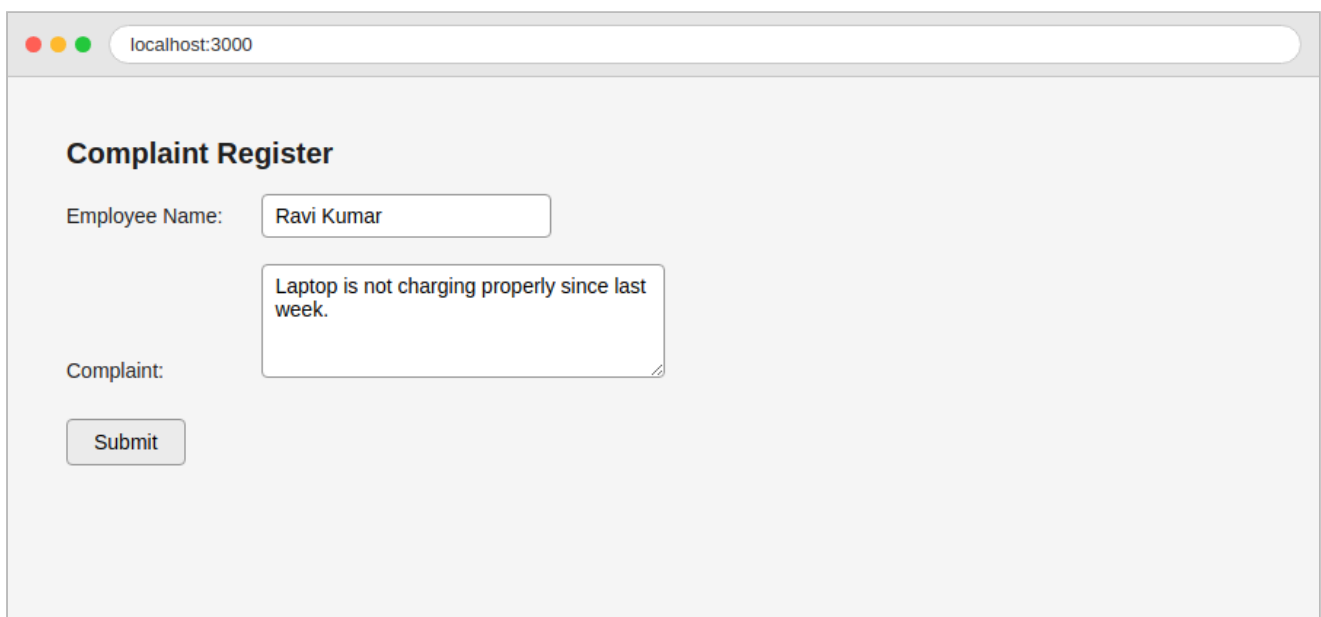
webpack compiled successfully
```

Output: Empty complaint form



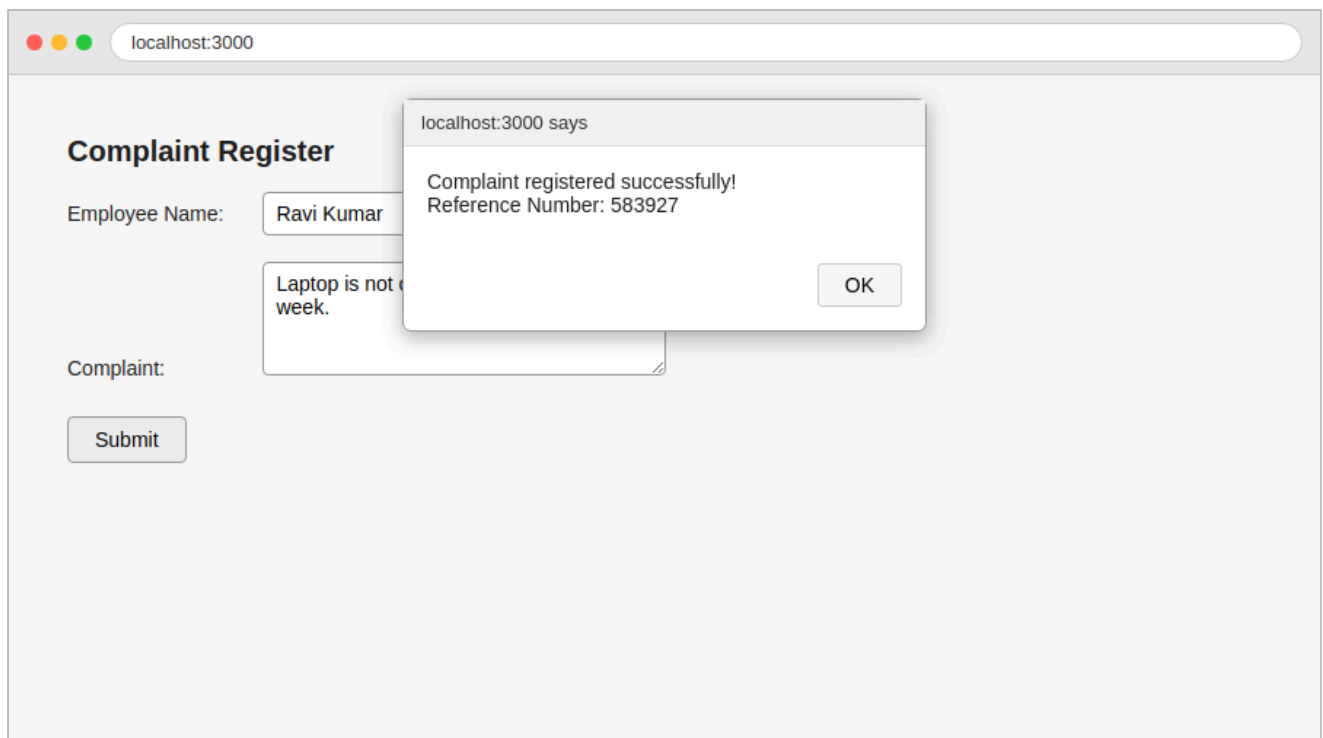
A screenshot of a web browser window with the address bar showing 'localhost:3000'. The page has a light gray background and a title 'Complaint Register'. Below the title, there is a form with two input fields. The first field is labeled 'Employee Name:' and is empty. The second field is labeled 'Complaint:' and is also empty. Below the 'Complaint:' field, there is a 'Submit' button.

Output: Form filled with employee name and complaint



A screenshot of a web browser window with the address bar showing 'localhost:3000'. The page has a light gray background and a title 'Complaint Register'. Below the title, there is a form with two input fields. The first field is labeled 'Employee Name:' and contains the text 'Ravi Kumar'. The second field is labeled 'Complaint:' and contains the text 'Laptop is not charging properly since last week.' Below the 'Complaint:' field, there is a 'Submit' button.

Output: Submit clicked – Reference number shown in alert



Result

The React form with controlled components (textbox and textarea) was implemented successfully. The handleSubmit event was used to generate and display a reference number for the complaint raised.